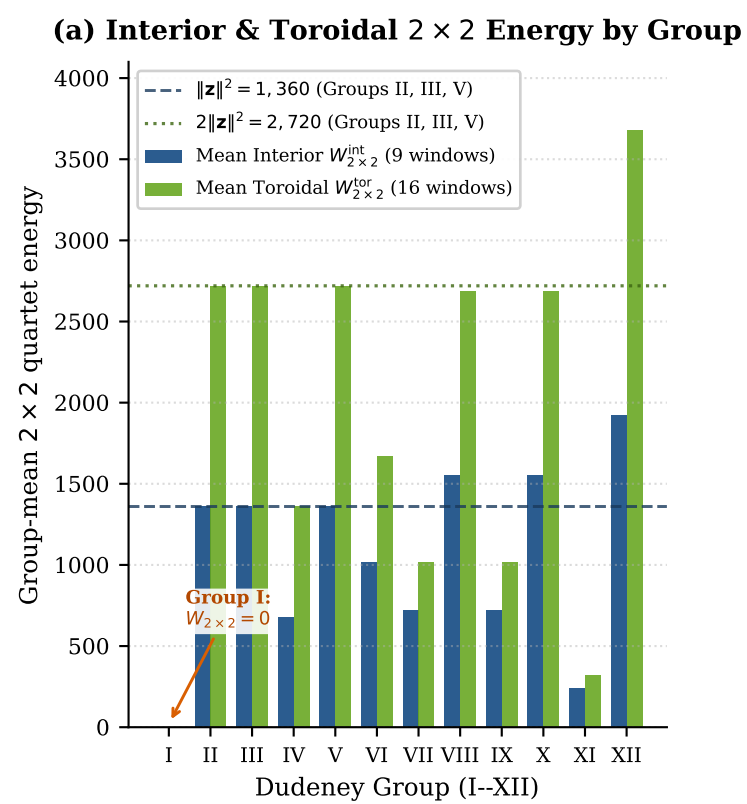
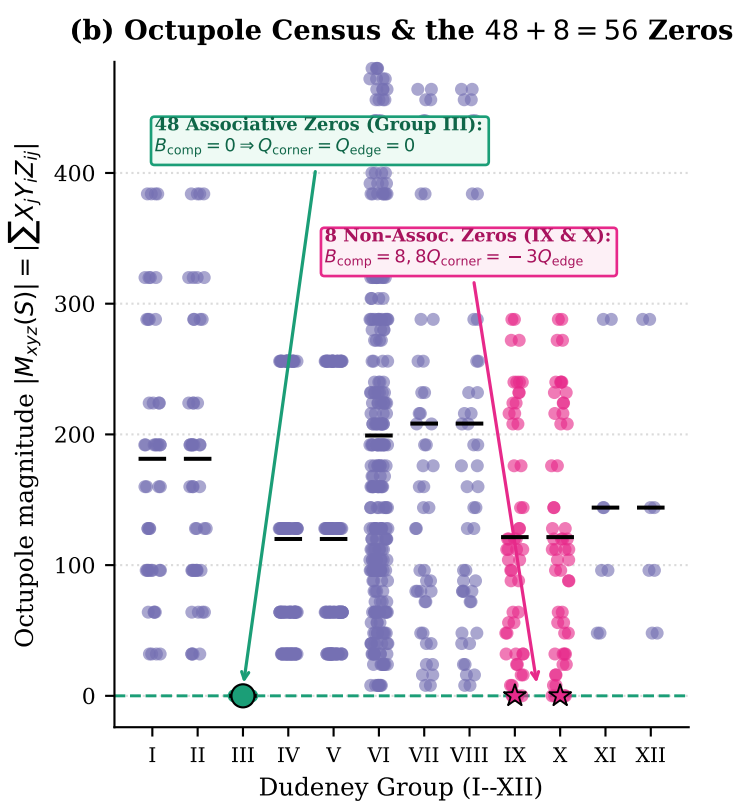
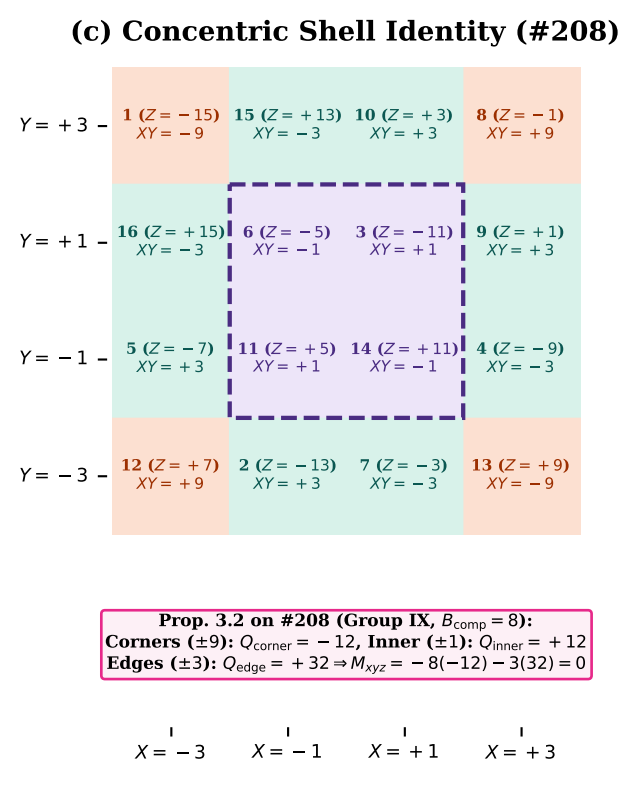

(a) Interior & Toroidal 2 × 2 Energy by Group
‖z‖² = 1, 360 (Groups II, III, V)
2‖z‖² = 2, 720 (Groups II, III, V)
Mean Interior W^int_{2×2} (9 windows)
Mean Toroidal W^tor_{2×2} (16 windows)
Group I:
W_{2×2} = 0
Group-mean 2 × 2 quartet energy
Dudeney Group (I--XII)
I II III IV V VI VII VIII IX X XI XII
0 500 1000 1500 2000 2500 3000 3500 4000


(b) Octupole Census & the 48 + 8 = 56 Zeros
48 Associative Zeros (Group III):
B_comp = 0 ⇒ Q_corner = Q_edge = 0
8 Non-Assoc. Zeros (IX & X):
B_comp = 8, 8Q_corner = −3Q_edge
Octupole magnitude |M_xyz(S)| = |∑ X_i Y_i Z_ij|
Dudeney Group (I--XII)
I II III IV V VI VII VIII IX X XI XII
0 100 200 300 400


(c) Concentric Shell Identity (#208)
Y = +3
Y = +1
Y = −1
Y = −3
1 (Z = −15) XY = −9
15 (Z = +13) XY = −3
10 (Z = +3) XY = +3
8 (Z = −1) XY = +9
16 (Z = +15) XY = −3
6 (Z = −5) XY = −1
3 (Z = −11) XY = +1
9 (Z = +1) XY = +3
5 (Z = −7) XY = +3
11 (Z = +5) XY = +1
14 (Z = +11) XY = −1
4 (Z = −9) XY = −3
12 (Z = +7) XY = +9
2 (Z = −13) XY = +3
7 (Z = −3) XY = −3
13 (Z = +9) XY = −9
X = −3
X = −1
X = +1
X = +3
Prop. 3.2 on #208 (Group IX, B_comp = 8):
Corners (±9): Q_corner = −12, Inner (±1): Q_inner = +12
Edges (±3): Q_edge = +32 ⇒ M_xyz = −8(−12) − 3(32) = 0